Seongmoon **Jeong**

PH.D. CANDIDATE

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I am a Ph.D. candidate in the Department of Artificial Intelligence at Sungkyunkwan University, advised by **Prof. Jong Hwan Ko**. My research focuses on **image compression for machine perception**, spanning low-level optimization and high-level tasks such as classification, object detection, and segmentation. I develop **learned, task-aware representations** that jointly optimize compression efficiency and downstream vision performance. More recently, I have investigated high-level vision directly in the **RAW domain**, before the image signal processing (ISP) pipeline, to preserve information relevant to machine perception. I have authored or co-authored **eight peer-reviewed publications**, including papers at CVPR, AVSS, ICCAD, and in IEEE TCAS-I, and contributed **four ISO/IEC MPEG VCM proposals**. My long-term goal is to enable **communication-efficient intelligent systems** for autonomous systems and edge-based machine vision. **Available for full-time roles starting March 2027.**

Education

Ph.D. in Artificial Intelligence (Expected February 2027) — *Suwon, Korea*

SUNGKYUNKWAN UNIVERSITY (SKKU) — *Mar. 2020 - Present*

- Received an Outstanding Scholarship awarded to promising students in the Department of Artificial Intelligence.

B.S. in Electronic and Electrical Engineering — *Suwon, Korea*

SUNGKYUNKWAN UNIVERSITY (SKKU) — *Mar. 2013 - Feb. 2020*

- GPA: 3.84/4.5 (Top 6.6%)

Research Experience

Graduate Researcher — *Suwon, Korea*

IRIS LAB, SUNGKYUNKWAN UNIVERSITY (SKKU) — *Mar. 2020 - Present*

- **Machine-Oriented Image Compression**: Codec-transferable and test-time adaptation across downstream vision tasks [**C4**, **C5**]
- **Task-Aware Preprocessing and JPEG Coding**: Adaptive downscaling, feature modulation, and region-based JPEG optimization [**C1**, **C3**, **J1**]
- **RAW-Domain Machine Vision**: Foundation-model-guided RGB-to-RAW generation for object detection [**C6**]

Project Experience

Video Coding for Machines — *Media Research Division, ETRI*

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU) — *2023 - 2024*

- Led research on codec-transferable adaptation for machine-oriented image compression [**C5**].

Adaptive and Efficient Preprocessing and Coding for Machine Vision Tasks and Input Images — *Media Research Division, ETRI*

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU) — *2022*

- Led research on adaptive image downscaling for rate-accuracy-latency optimization [**C3**].

Multipurpose Visual Information Compression for Human and Machine Vision — *Ministry of Science and ICT (MSIT), Korea*

PARTICIPATING RESEARCHER (BASIC RESEARCH LABORATORY, IRIS LAB., SKKU) — *2025 - Present*

- Contributing to unified visual compression methods supporting both human viewing and machine perception.

Selected Publications

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection — *IEEE AVSS*

Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko — *Sep. 2026*

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression — *IEEE AVSS*

Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko — *Sep. 2026*

C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability — *IEEE/CVF CVPR*

Un Ki Park, **Seongmoon Jeong**, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko — *Jun. 2025*

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression — *IEEE AICAS*

Hangyul Choi, **Seongmoon Jeong**, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko — *Apr. 2024*

Standardization Activities

[VCM] Neural network based post-filter for VCM *Geneva*
ISO/IEC JTC1 / SC29 / WG4 M71305 *Jan. 2025*

[VCM] Neural network based pre-filter for VCM *Geneva*
ISO/IEC JTC1 / SC29 / WG4 M71304 *Jan. 2025*

[VCM] Learned joint filter network for VCM *Kemer*
ISO/IEC JTC1 / SC29 / WG4 M69990 *Nov. 2024*

[VCM] Learned pre-filter network for VCM *Sapporo*
ISO/IEC JTC1 / SC29 / WG4 M69085 *Jan. 2024*

Honors & Awards

2024	**3rd Place**, DAC System Design Contest - GPU Track	*San Francisco, USA*
2020	**2nd Place**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT	*Korea*
2019	**Honorable Mention**, Big Data Center Open Innovation Challenge	*Seongnam, Korea*
2019	**Outstanding Work Award**, Capstone Design Project, SKKU	*Suwon, Korea*
2018	**Dean's List Award**, Fall semester, SKKU	*Suwon, Korea*
2017	**Dean's List Award**, Fall semester, SKKU	*Suwon, Korea*

Full Publications

INTERNATIONAL CONFERENCE

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection *IEEE AVSS*
Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko *Sep. 2026*

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression *IEEE AVSS*
Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko *Sep. 2026*

C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability *IEEE/CVF CVPR*
Un Ki Park, **Seongmoon Jeong**, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko *Jun. 2025*

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression *IEEE AICAS*
Hangyul Choi, **Seongmoon Jeong**, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko *Apr. 2024*

C2 Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays *IEEE/ACM ICCAD*
Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, **Seongmoon Jeong**, Jong Hwan Ko *Oct. 2023*

C1 Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation *IEEE ICME Workshops*
Seongmoon Jeong, Kang Eun Jeon, Jong Hwan Ko *Jul. 2023*

INTERNATIONAL JOURNAL

J2 KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping *IEEE TCAS-I*
Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, **Seongmoon Jeong**, Jong Hwan Ko *Dec. 2024*

J1 An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression *Pattern Recognition Letters*
Seonghye Jeong, **Seongmoon Jeong**, Simon S. Woo, Jong Hwan Ko *Jan. 2023*

Research Engineering Experience

ML Systems & Research Infrastructure

IRIS LAB., SUNGKYUNKWAN UNIVERSITY

- **Multi-node experiment orchestration:** Built a Python-based multi-node experiment orchestration pipeline across four servers with 16 RTX 2080 Ti GPUs, spawning one process per GPU, sharding training and evaluation datasets across workers, and aggregating outputs on a head node for downstream evaluation [**C4**].
- **Input pipeline optimization:** Profiled data-loading and augmentation bottlenecks and optimized pipelines using TensorFlow Datasets and Grain, reducing input stalls during neural network experiments.
- **Reproducible experimentation:** Standardized environments and automated workflows using Docker, Conda/Mamba, uv, and Pixi, improving consistency across research machines.

Student Mentoring

Hangyul Choi

M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY, NOW AT MX BUSINESS, SAMSUNG ELECTRONICS *2023 - 2024*

- Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression [**C3**]

Chanung Park

GRADUATE STUDENT AT SUNGKYUNKWAN UNIVERSITY *2024 - Present*

- Mentored and collaborated on research in Video Coding for Machines (VCM) and multipurpose visual information compression through the BRL project.

Skills

Programming Languages	Python, Shell scripting, C/C++
ML Frameworks	PyTorch, TensorFlow, JAX, TensorRT
Development Frameworks	Docker, Conda, uv

Reference

Jong Hwan Ko

Ph.D. advisor

ASSOCIATE PROFESSOR *Dept. of ECE, SKKU, Suwon, Korea*

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